



Sealed Pick

A party game for three on Midnight. Everyone seals a pick through a Compact contract; the seals break together.

THE PROBLEM

Hidden, simultaneous picks need a referee.

Coordination games, blind votes and sealed guesses only work when two rules hold:

- Nobody sees another pick before choosing.
- Nobody changes a pick after seeing someone else's.

Online, a server usually holds every pick, and everyone has to trust it.

Three players, three seals, one reveal.

01

Seal

Each phone keeps its choice and a random nonce in private state. The commit circuit proves the player owns the seat and the choice is in range, and discloses only the seal.

02

Wait

The public ledger shows three seals and a counter. No choice exists on chain yet, so there is nothing to peek at.

03

Break

After the last seal, the openings go to one player. The reveal circuit checks all three against their seals and publishes the choices and the score in one transaction.

THE LIVE TABLE

Play a round in the browser.

Three phones and the public ledger, in one tab. The compiled Compact circuits run in the page.

After a seal, "Cheat with a modified app" sends a second pick straight to the circuit. It answers "Player already committed".

luoy16002-svg.github.io/sealed-pick

**Sealed Pick**
Three players. One question. No talking.

● Simulator

the compiled Compact circuits run in this tab, no proofs

🔗 Source

ROUND 1

NO MATCHES YET

You lost each other in a huge train station. Where do you wait?

Everyone seals a pick. When all three seals are on the ledger, they break together. You win if all three picked the same thing.

▶ Play a round for me

Amaec9b...b55c

REFUSED

4A3A

Sealed. The ledger holds only this seal.

ONLY ON AMA'S PHONE
Under the big clock
plus a random 32-byte nonce, in private state

The circuit refused the swap
A modified app skipped its own checks and sent a second pick for this seat. The circuit answered "Player already committed".

Lior7d44...69d2

9269

Sealed. The ledger holds only this seal.

ONLY ON LIOR'S PHONE
Under the big clock
plus a random 32-byte nonce, in private state

 Cheat with a modified app

Mei9a61...68cf

YOUR PICK · ONLY YOU SEE IT

Under the big clock✓

Main entrance☐

Ticket hall☐

Platform 1☐

Your choice stays on this phone.
The ledger only gets a seal.

Seal my pick

Public ledger
What everyone can read, the server included
room f8265be314...c56505 · 4 options

Lobby	Sealing	All sealed	Revealed
● ec9b46...b55c	● 4a3ad66209...83ab0d		
● 7d441e...69d2	● 92696e1a7d...e6a5cd		
● 9a6127...68cf		no seal yet	

Names stay on the phones. The ledger only sees seat pseudonyms and seals.

1 seal to go

CONTRACT CALLS

commit	seat 0 · forged swap refused: Player already committed	19:35:58
commit	seat 1 · seal 92696e1a...a5cd	19:35:58
commit	seat 0 · seal 4a3ad662...ab0d	19:35:57
joinRoom	seat 2 taken · lobby closed	19:35:56
joinRoom	seat 1 taken	19:35:56

What the ledger sees.

On each phone, private state

- The player's random 32-byte secret, which proves seat ownership
- The choice and its nonce until the reveal

On the public ledger

- Room ID, option count and phase
- Seat pseudonyms and seals
- After the reveal: the three choices and the score

```
pseudonym = persistentHash(domain, roomId, secret)
seal = persistentCommit((domain, roomId, seat, choice), nonce)
```

BUILT AND CHECKED

Running, tested, proven on a devnet.

Compact 0.31.1

Four circuits compile with proving and verifying keys.

33 tests pass

Rules, every commit order, all 64 answer triples, and a raw-ledger check that no choice appears early.

Browser table

Runs the generated circuits in the page. A forged second commitment is refused by the circuit.

```
npm run devnet:e2e · local devnet · real proofs
```

deploy	block 8	36a39a1d...ef2c	SucceedEntirely
createRoom	block 11	afa72cfe...ca7d	SucceedEntirely
joinRoom	block 14	7741716e...a0db	SucceedEntirely
joinRoom	block 17	530b7dbe...e3e9	SucceedEntirely
commit	block 20	b49299f0...4125	SucceedEntirely
commit	block 23	8d93f834...9295	SucceedEntirely
commit	block 26	401e58f3...6461	SucceedEntirely
reveal	block 30	b1420b94...3da9	SucceedEntirely

NEXT

Where Sealed Pick goes from here.

Who plays

Friends on a call, classrooms learning about focal points, stream audiences voting without seeing the running tally.

What we build next

The same round on Preprod: the script is ready and waits for faucet funds. Openings sent between phones over an encrypted channel. Room links and more seats.

How it pays

Free to play. Paid question packs, and branded rooms for streamers and events.

TRY IT

Three phones, one ledger, no peeking.

```
luoy16002-svg.github.io/sealed-pick
```

Press “Play a round for me”, or play all three phones yourself. On a phone, the table shows one seat at a time.

Locally: `npm ci && npm run build && npm run web:dev`, then open `localhost:5391`.